# Solution Requirements (Functional & Non-functional)

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| **Date** | 28 October 2026 |
| **Team ID** | EN25CS3010362 |
| **Project Name** | Formula 1 Race Intelligence Dashboard |
| **Maximum Marks** | As per supplied template |

## Functional Requirements

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task) | Priority |
|---|---|---|---|
| FR-1 | Data Access & Integration | Use the Formula 1 dataset connected to the Tableau workbook. Relevant fields include race records, seasons, circuits/host locations, drivers, teams, results/points, tyre compounds, tyre age, and retirement reasons where available. | High |
| FR-2 | Data Processing | Use suitable field types and consistent filters/aggregations. Check missing values, duplicate records, and category labels before interpreting charts. Document only data-cleaning steps actually performed. | High |
| FR-3 | Business Analysis | Display coverage KPIs (race records, seasons, circuits); compare driver and team performance; summarize tyre-compound usage; examine podium achievements, reliability/DNF reasons, and season-wise measures where supported. | High |
| FR-4 | Business Insights & Analytical Outputs | Provide KPI cards, driver leaderboard, tyre usage, team ranking, podium view, reliability/DNF view, host-country map, completion-rate view, season-wise team performance, tyre usage by season, and a guided Tableau Story. | High |
| FR-5 | User Interaction & Analysis | Allow users to use available Season, Team, Tyre Compound, and DNF Reason filters; compare driver/team results and inspect chart marks. Confirm filter scope and aggregation in Tableau. | High |
| FR-6 | Data Update & Reporting | Publish the workbook and Story to Tableau Public and link them from the Flask website. Republish when the underlying workbook/data is updated; an automatic refresh schedule has not been verified. | Medium |

## Non-functional Requirements

| NFR No. | Requirement | Description |
|---|---|---|
| NFR-1 | Usability | Use descriptive titles, readable labels and legends, consistent terminology, and a simple |

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| | | website entry point with Dashboard and Story links. |
| NFR-2 | Security | Keep passwords, API keys, and other secrets out of the public repository. Since the site and dashboard are public, do not expose private data. |
| NFR-3 | Reliability | Verify website, dashboard, Story, and navigation links before demonstration. Keep direct Tableau URLs as fallbacks. |
| NFR-4 | Performance | Measure website load, dashboard rendering, and filter response with repeatable tests if performance claims are required. No verified timings were supplied. |
| NFR-5 | Availability | Website is deployed on Render; dashboard and Story are hosted on Tableau Public. Availability depends on those services and internet connectivity. |
| NFR-6 | Scalability | For larger datasets/audiences, review workbook complexity, data refresh strategy, hosting capacity, and whether a managed data platform is needed. No load-test evidence was supplied. |

## Project Links

Website: https://f1-race-intelligence-mhgm.onrender.com/

GitHub: https://github.com/gourisanjaysharma-source/F1-Race-Intelligence

Tableau Dashboard:
https://public.tableau.com/app/profile/gouri.sharma8417/viz/f1tableauproject/F1RACEINTELLIGENCEDASHBOARD

Tableau Story: https://public.tableau.com/views/f1tableauproject/Formula1RaceIntelligence